

Data Privacy: Homeworks #4

Due on Wednesday 11PM, June 17, 2026

Note: Submit homework via LearnUS. Topics: membership inference attacks and machine unlearning. Throughout this homework, you may assume that all random variables you encounter have continuous densities on $\mathbb{R}$ (or $\mathbb{R}^d$) and that all integrals are well-defined. No measure theory is required.

1. **Neyman–Pearson Lemma.** Let T be a real-valued random variable. Under hypothesis H_b (for $b \in \{0, 1\}$), assume T has a continuous density p_b on $\mathbb{R}$, with $p_0(t) > 0$ for every t . Define the *likelihood ratio*

$$\Lambda(t) = \frac{p_1(t)}{p_0(t)}.$$

A *test* is a function $\phi : \mathbb{R} \rightarrow [0, 1]$, where $\phi(t)$ is the probability of rejecting H_0 after observing $T = t$. Define

$$\alpha(\phi) = \mathbb{E}_{H_0}[\phi(T)] \quad (\text{size, type-I error}), \quad \beta(\phi) = \mathbb{E}_{H_1}[\phi(T)] \quad (\text{power}).$$

Fix a level $\alpha_0 \in (0, 1)$, and *assume that the function $G(\tau) := \Pr_{H_0}[\Lambda(T) > \tau]$ is continuous on $[0, \infty)$* . (This holds automatically in the Gaussian models you will see in Problems 3, 5, and 6.) Prove the following three claims.

- (a) **Existence.** Show that there exists a threshold $\tau^* \geq 0$ such that the test

$$\phi^*(t) = \mathbf{1}[\Lambda(t) > \tau^*]$$

satisfies $\alpha(\phi^*) = \alpha_0$ exactly.

Hint: Show that G is non-increasing on $[0, \infty)$ with $G(0) \leq 1$ and $G(\tau) \rightarrow 0$ as $\tau \rightarrow \infty$. Then apply the intermediate value theorem.

- (b) **Optimality.** Show that for every test ϕ with $\alpha(\phi) \leq \alpha_0$, we have $\beta(\phi) \leq \beta(\phi^*)$.

Hint: Verify that for every t , $(\phi^*(t) - \phi(t))(p_1(t) - \tau^* p_0(t)) \geq 0$, then integrate over $\mathbb{R}$.

- (c) **Uniqueness.** Suppose ϕ is any most-powerful level- α_0 test (i.e., $\alpha(\phi) \leq \alpha_0$ and $\beta(\phi) = \beta(\phi^*)$). Show that $\phi(t) = \phi^*(t)$ for every t with $\Lambda(t) \neq \tau^*$, except possibly on a set of probability 0.

Hint: The integrand from part (b) is non-negative and now integrates to 0.

2. **DP $\Rightarrow$ MIA Bounds.** Let $M : \mathcal{X}^n \rightarrow \mathcal{Y}$ be (ε, δ) -differentially private, and let D_0, D_1 be two neighboring datasets. Consider any (possibly randomized) binary test $\mathcal{A} : \mathcal{Y} \rightarrow \{0, 1\}$ that tries to distinguish $M(D_0)$ from $M(D_1)$. Write

$$\text{FPR}(\mathcal{A}) = \Pr[\mathcal{A}(M(D_0)) = 1], \quad \text{TPR}(\mathcal{A}) = \Pr[\mathcal{A}(M(D_1)) = 1].$$

- (a) Show that

$$\text{TPR}(\mathcal{A}) \leq e^\varepsilon \text{FPR}(\mathcal{A}) + \delta, \quad 1 - \text{FPR}(\mathcal{A}) \leq e^\varepsilon (1 - \text{TPR}(\mathcal{A})) + \delta.$$

- (b) Define the MI advantage $\text{Adv}(\mathcal{A}) = \text{TPR}(\mathcal{A}) - \text{FPR}(\mathcal{A})$. Show the loose bound

$$\text{Adv}(\mathcal{A}) \leq e^\varepsilon - 1 + \delta.$$

Then, by combining both inequalities of part (a), derive the tighter Kairouz–Oh–Viswanath bound

$$\text{Adv}(\mathcal{A}) \leq \frac{e^\varepsilon - 1 + 2\delta}{e^\varepsilon + 1}.$$

- (c) Conclude that for $\varepsilon = 0.1$ and $\delta = 10^{-5}$, no adversary can exceed advantage ≈ 0.105 , but for $\varepsilon = 8$ the bound is vacuous.

3. **Bayes-Optimal Membership Test.** Let $B \in \{0, 1\}$ be a Bernoulli(π) membership indicator, and let T be an observed statistic with continuous density $p_b(t)$ given $B = b$. Consider any decision rule $\hat{B} : \mathbb{R} \rightarrow \{0, 1\}$, and measure its quality by the 0-1 error $\Pr[\hat{B}(T) \neq B]$.

- (a) Compute the posterior $\Pr[B = 1 \mid T = t]$ in terms of p_0, p_1, π .
(b) Show that the Bayes-optimal rule (the MAP rule) is

$$\hat{B}_{\text{MAP}}(t) = \mathbf{1}\left[\Lambda(t) \geq \frac{1 - \pi}{\pi}\right], \quad \Lambda(t) = \frac{p_1(t)}{p_0(t)},$$

and connect this to Problem 1: the Bayes-optimal MIA is a likelihood-ratio test, and only the threshold changes with the prior.

- (c) Suppose $T = \ell(f_\theta, z)$ is the loss of model f_θ on the queried point z , and that $\ell \mid B = 1 \sim \mathcal{N}(\mu_{\text{in}}, \sigma^2)$, $\ell \mid B = 0 \sim \mathcal{N}(\mu_{\text{out}}, \sigma^2)$ with $\mu_{\text{in}} < \mu_{\text{out}}$. Show that the optimal rule reduces to a single-threshold loss test $\hat{B} = \mathbf{1}[\ell \leq c]$ and compute c explicitly when $\pi = 1/2$.

4. **Generalization Gap as a Leakage Lower Bound (Yeom).** Suppose the loss is bounded: $\ell(f_\theta, z) \in [0, B]$ for all θ, z . Define

$$R_{\text{train}} = \mathbb{E}_{z \sim \text{Unif}(D)}[\ell(f_\theta, z)], \quad R_{\text{pop}} = \mathbb{E}_{z \sim \mathcal{P}}[\ell(f_\theta, z)], \quad \Delta = R_{\text{pop}} - R_{\text{train}} \geq 0.$$

The MIA game: sample $b \sim \text{Unif}\{0, 1\}$. If $b = 1$, return z uniform over D ; otherwise, return $z \sim \mathcal{P}$. The adversary uses a threshold attack $\mathcal{A}_\tau(z) = \mathbf{1}[\ell(f_\theta, z) \leq \tau]$. Let $F_1(\tau) = \Pr[\ell \leq \tau \mid b = 1]$ and $F_0(\tau) = \Pr[\ell \leq \tau \mid b = 0]$; assume both have continuous densities on $[0, B]$.

- (a) Show that the advantage of $\mathcal{A}_\tau$ is

$$\text{Adv}(\mathcal{A}_\tau) = F_1(\tau) - F_0(\tau).$$

- (b) Prove the *layer-cake identity*: for any random variable X with continuous density on $[0, B]$,

$$\mathbb{E}[X] = \int_0^B (1 - F_X(t)) dt.$$

Then deduce

$$\int_0^B (F_1(t) - F_0(t)) dt = \Delta.$$

- (c) Conclude that there exists a threshold $\tau^* \in [0, B]$ for which the threshold attack achieves advantage

$$\text{Adv}(\mathcal{A}_{\tau^*}) \geq \frac{\Delta}{B}.$$

In words: the generalization gap is a *lower* bound on the leakage of the best threshold attack — not an upper bound, contrary to a common informal slogan.

Hint: $\Delta = \int_0^B (F_1 - F_0) dt \leq B \cdot \sup_t (F_1(t) - F_0(t))$.

- (d) Now suppose additionally that under each hypothesis the loss is Gaussian:

$$\ell \mid b = 1 \sim \mathcal{N}(R_{\text{train}}, \sigma^2), \quad \ell \mid b = 0 \sim \mathcal{N}(R_{\text{pop}}, \sigma^2).$$

(Ignore the issue that Gaussians are not literally bounded; the formula below is the correct one for the full range $\mathbb{R}$.) Show that the maximum-advantage threshold attack has $\tau^* = (R_{\text{train}} + R_{\text{pop}})/2$ and achieves

$$\text{Adv}^* = 2\Phi\left(\frac{\Delta}{2\sigma}\right) - 1.$$

For small Δ/σ , derive the linear approximation $\text{Adv}^* \approx \Delta/(\sigma\sqrt{2\pi})$ and explain why noisier loss distributions reduce leakage.

5. **LiRA Closed Form and Decision Boundary.** Assume that for a target record z , the per-shadow-model loss has Gaussian distributions under the two membership hypotheses:

$$\ell \mid (z \in D) \sim \mathcal{N}(\mu_{\text{in}}, \sigma_{\text{in}}^2), \quad \ell \mid (z \notin D) \sim \mathcal{N}(\mu_{\text{out}}, \sigma_{\text{out}}^2).$$

- (a) Derive the log-likelihood ratio

$$\log \Lambda(\ell) = \log \frac{\mathcal{N}(\ell; \mu_{\text{in}}, \sigma_{\text{in}}^2)}{\mathcal{N}(\ell; \mu_{\text{out}}, \sigma_{\text{out}}^2)} = \frac{(\ell - \mu_{\text{out}})^2}{2\sigma_{\text{out}}^2} - \frac{(\ell - \mu_{\text{in}})^2}{2\sigma_{\text{in}}^2} + \log \frac{\sigma_{\text{out}}}{\sigma_{\text{in}}}.$$

- (b) Show that for general $\sigma_{\text{in}} \neq \sigma_{\text{out}}$, the optimal decision region $\{\log \Lambda > \log \tau\}$ is one of: a bounded interval, the complement of a bounded interval, a half-line, all of $\mathbb{R}$, or empty. Identify which case occurs based on the sign of $\sigma_{\text{in}} - \sigma_{\text{out}}$.
- (c) Show that in the homoscedastic case $\sigma_{\text{in}} = \sigma_{\text{out}} = \sigma$, the optimal MIA reduces to the single threshold $\mathcal{A}(z) = \mathbf{1}[\ell \leq c]$, and express c in terms of $\mu_{\text{in}}, \mu_{\text{out}}, \sigma, \tau$.
- (d) In the homoscedastic case, derive the closed-form ROC: at false-positive rate α , the optimal true-positive rate is

$$\text{TPR}(\alpha) = \Phi\left(\Phi^{-1}(\alpha) + \frac{\mu_{\text{out}} - \mu_{\text{in}}}{\sigma}\right),$$

where Φ is the standard normal CDF.

6. **Concrete MIA on a Gaussian Mean Model.** A “training” algorithm maps $D = (z_1, \dots, z_n)$ with $z_i \in \mathbb{R}$ to the parameter

$$\theta(D) = \bar{z} + W, \quad \bar{z} = \frac{1}{n} \sum_{i=1}^n z_i, \quad W \sim \mathcal{N}(0, \rho^2) \text{ independent of } D.$$

Assume the population $\mathcal{P}$ is $\mathcal{N}(0, 1)$, and the adversary observes θ plus a candidate point z^* . The MIA game: $b \sim \text{Unif}\{0, 1\}$. If $b = 1$, $z^* = z_1$ (a training sample, included in D); if $b = 0$, $z^* \sim \mathcal{P}$ independent of D .

- (a) Show that conditional on z^* ,

$$\theta \mid (b = 1) \sim \mathcal{N}\left(\frac{z^*}{n}, \frac{n-1}{n^2} + \rho^2\right), \quad \theta \mid (b = 0) \sim \mathcal{N}\left(0, \frac{1}{n} + \rho^2\right).$$

- (b) Show that the optimal adversary’s test statistic is the (conditional) likelihood ratio $\Lambda(\theta, z^*) = p(\theta \mid b = 1, z^*)/p(\theta \mid b = 0, z^*)$, and write down $\log \Lambda$ as a quadratic in θ (with z^* treated as a known constant).
(c) Now suppose the inputs are clipped, $|z_i| \leq 1$, so the L_2 -sensitivity of $\bar{z}$ under one-element neighbors is at most $2/n$. The Gaussian mechanism with

$$\rho \geq \frac{2}{n\varepsilon} \sqrt{2 \log(1.25/\delta)}$$

makes the release (ε, δ) -DP. Using Problem 2(b), give a numerical upper bound on the optimal MIA advantage for $\varepsilon = 0.1$, $\delta = 10^{-5}$, and comment on why $\varepsilon = 1$ already gives a vacuous bound.

7. **Influence-Function Unlearning (Closed Form).** Let $L(\theta; D) = \frac{1}{n} \sum_{i=1}^n \ell(\theta; z_i)$ be a twice-differentiable, λ -strongly convex loss in $\theta \in \mathbb{R}^d$. Recall that λ -strong convexity means the Hessian $H_\theta = \nabla^2 L(\theta; D)$ satisfies $H_\theta \succeq \lambda I$, which in turn implies H_θ is invertible with $\|H_\theta^{-1}\|_{\text{op}} \leq 1/\lambda$. Let $\theta_o = \arg \min_\theta L(\theta; D)$ and $\theta_{-z} = \arg \min_\theta L(\theta; D \setminus \{z\})$.

- (a) Consider the parametric family

$$L_t(\theta) = L(\theta; D) - \frac{t}{n} \ell(\theta; z), \quad t \in [0, 1],$$

and let $\theta(t) = \arg \min_\theta L_t(\theta)$. (Note that $L_0 = L(\cdot; D)$ gives $\theta(0) = \theta_o$, and L_1 is a positive scalar multiple of $L(\cdot; D \setminus \{z\})$, so $\theta(1) = \theta_{-z}$.) Use the first-order optimality condition $\nabla_\theta L_t(\theta(t)) = 0$ and differentiate with respect to t (the *implicit function theorem*) to derive

$$\left. \frac{d\theta(t)}{dt} \right|_{t=0} = \frac{1}{n} H_{\theta_o}^{-1} \nabla_\theta \ell(\theta_o; z).$$

- (b) Conclude the first-order Newton-step approximation

$$\theta_{-z} - \theta_o \approx \frac{1}{n} H_{\theta_o}^{-1} \nabla_\theta \ell(\theta_o; z).$$

The “unlearned” parameter is then $\theta_u(D, z) = \theta_o + \frac{1}{n} H_{\theta_o}^{-1} \nabla_\theta \ell(\theta_o; z)$.

- (c) Assume in addition that ℓ is G -Lipschitz in θ (so $\|\nabla_{\theta}\ell(\theta; z)\| \leq G$ for all θ, z). Show that the L_2 -sensitivity of the unlearned parameter to the choice of deleted point is bounded:

$$\sup_{z, z'} \|\theta_u(D, z) - \theta_u(D, z')\| \leq \frac{2G}{\lambda n}.$$

(Treat $H_{\theta_o}^{-1}$ as fixed; its operator norm is $\leq 1/\lambda$ by strong convexity. This is the first-order sensitivity used in certified unlearning.)

8. **Certified (ε, δ) -Unlearning by Gaussian Noise.** Building on Problem 7, let $A(D) = \arg \min_{\theta} L(\theta; D)$ be the (deterministic) full-retrain map, and define the noisy first-order unlearning algorithm

$$\mathcal{U}(D, z) = \theta_o(D) + \frac{1}{n} H_{\theta_o(D)}^{-1} \nabla_{\theta} \ell(\theta_o(D); z) + N, \quad N \sim \mathcal{N}(0, \sigma^2 I_d).$$

Assume ℓ is λ -strongly convex and G -Lipschitz, and that the first-order approximation in Problem 7(b) has L_2 error bounded by

$$\eta := \|\theta_u^{1st} - A(D \setminus \{z\})\| \leq \frac{C_2 G^2}{\lambda^2 n^2}$$

for some universal constant C_2 (a known second-order bound that you may use without proof).

- (a) Show that the deterministic part of $\mathcal{U}(D, z)$ differs from $A(D \setminus \{z\})$ in L_2 norm by at most

$$\eta \leq \frac{C_2 G^2}{\lambda^2 n^2}.$$

- (b) Recall the Gaussian-mechanism privacy bound: if $f, g : \mathcal{X}^n \rightarrow \mathbb{R}^d$ satisfy $\|f(X) - g(X)\| \leq \eta$ for every X , then $f(X) + N$ and $g(X) + N$ are (ε, δ) -indistinguishable provided

$$\sigma \geq \frac{\eta}{\varepsilon} \sqrt{2 \log(1.25/\delta)}.$$

Apply this with $f = \theta_u^{1st}$ and $g = A(\cdot \setminus \{z\})$ to conclude that

$$\sigma \geq \frac{C_2 G^2}{\lambda^2 n^2 \varepsilon} \sqrt{2 \log(1.25/\delta)}$$

yields an (ε, δ) -certified unlearning algorithm.

- (c) How does the required noise scale with n ? Compare to the Gaussian-mechanism noise needed for (ε, δ) -DP of the trained model itself ($\sigma \propto 1/(n\varepsilon)$). Explain in one or two sentences why the unlearning noise is smaller for large n .

9. **NPO Has a Bounded Gradient; GA Does Not.** For LLM unlearning, let $\pi_{\theta}(y \mid x)$ denote the model's likelihood of completion y given prompt x , and let π_{ref} be a fixed reference model. The forget set is $\mathcal{D}_F$. Two unlearning losses:

$$\begin{aligned} \mathcal{L}_{\text{GA}}(\theta) &= -\mathbb{E}_{(x,y) \sim \mathcal{D}_F} [-\log \pi_{\theta}(y \mid x)] = \mathbb{E}[\log \pi_{\theta}(y \mid x)], \\ \mathcal{L}_{\text{NPO}}(\theta) &= \frac{2}{\beta} \mathbb{E} \left[\log \left(1 + \left(\frac{\pi_{\theta}(y \mid x)}{\pi_{\text{ref}}(y \mid x)} \right)^{\beta} \right) \right] \quad (\beta > 0). \end{aligned}$$

- (a) Compute $\nabla_{\theta} \mathcal{L}_{\text{GA}}$. Using the identity $\nabla_{\theta} \log \pi_{\theta} = \nabla_{\theta} \pi_{\theta} / \pi_{\theta}$, show that for any sample with $\pi_{\theta}(y \mid x) \rightarrow 0$ (and $\|\nabla_{\theta} \pi_{\theta}\|$ bounded away from 0 in some direction), $\|\nabla_{\theta} \mathcal{L}_{\text{GA}}\| \rightarrow \infty$. Explain in one sentence why this drives the GA optimization to escape the data distribution (“GA collapse”).
- (b) Let $u(\theta) = \beta \log(\pi_{\theta}(y \mid x) / \pi_{\text{ref}}(y \mid x))$. Show that

$$\mathcal{L}_{\text{NPO}}(\theta) = \frac{2}{\beta} \mathbb{E}[\log(1 + e^{u(\theta)})], \quad \frac{\partial \mathcal{L}_{\text{NPO}}}{\partial u} = \frac{2}{\beta} \sigma(u) \in \left(0, \frac{2}{\beta}\right),$$

where σ is the logistic sigmoid. Conclude that

$$\|\nabla_{\theta} \mathcal{L}_{\text{NPO}}\| \leq 2 \mathbb{E}[\|\nabla_{\theta} \log \pi_{\theta}(y \mid x)\|].$$

- (c) Show that the right-hand bound above is uniform in θ : in particular, even as $\pi_{\theta}(y \mid x) \rightarrow 0$, the per-sample contribution $\sigma(u) \nabla_{\theta} \log \pi_{\theta}$ to the NPO gradient does *not* blow up, in sharp contrast to part (a). State briefly why this bounded-per-step divergence is what makes NPO “safe” for unlearning.

10. **Black-Box Unlearning Audits Require Many Queries.** Let θ_o and θ_u be two LLMs with output distributions P_0 and P_1 on a fixed prompt distribution. For two distributions P, Q on the same sample space, define the *total variation distance*

$$\text{TV}(P, Q) = \sup_A |P(A) - Q(A)|,$$

where the supremum is over events A . Assume $\text{TV}(P_0, P_1) \leq \eta$. An auditor draws q i.i.d. prompts $x_1, \dots, x_q$, observes one sample $y_i \sim P_b(\cdot \mid x_i)$ per prompt, and outputs $\hat{b} \in \{0, 1\}$. Write $P_b^{\otimes q}$ for the product distribution of the q observations under b .

- (a) **(Le Cam two-point lower bound.)** Show that for any test based on q samples,

$$\Pr_{b=0}[\hat{b} \neq 0] + \Pr_{b=1}[\hat{b} \neq 1] \geq 1 - \text{TV}(P_0^{\otimes q}, P_1^{\otimes q}).$$

Hint: Identify $\hat{b}$ with a function $\psi : \mathcal{X}^q \rightarrow [0, 1]$ (the probability of declaring $b = 1$). Then prove the auxiliary fact: for any distributions P, Q and any $\psi : \mathcal{X}^q \rightarrow [0, 1]$, $\mathbb{E}_P[\psi] - \mathbb{E}_Q[\psi] \leq \text{TV}(P, Q)$.

(For the auxiliary fact, write $\psi(x) = \int_0^1 \mathbf{1}[\psi(x) > s] ds$ and integrate.)

- (b) Show that $\text{TV}(P_0^{\otimes q}, P_1^{\otimes q}) \leq q \cdot \text{TV}(P_0, P_1) \leq q\eta$, by a hybrid (telescoping) argument: define $H_k = P_0^{\otimes k} \otimes P_1^{\otimes (q-k)}$ for $k = 0, \dots, q$, and bound $\text{TV}(H_k, H_{k-1})$.
- (c) Conclude that to guarantee total error $\leq 2\alpha$ (i.e., $\Pr_0[\hat{b} \neq 0] + \Pr_1[\hat{b} \neq 1] \leq 2\alpha$), the auditor needs

$$q \geq \frac{1 - 2\alpha}{\eta}.$$

Tighter χ^2 -style bounds give $q = \Omega(1/\eta^2)$. Why does this make single-prompt audits of LLM unlearning fundamentally insufficient when θ_u, θ_o differ only by a small distributional perturbation?